# Compact Academic Assistant via DPO, MCQA, Quantization, and RAG

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## 1 Introduction

We propose a compact academic assistant optimized for EPFL students using a four-stage pipeline: (1) supervised fine-tuning for multiple-choice QA (MCQA), (2) Direct Preference Optimization (DPO) for alignment, (3) quantization for efficiency, and (4) Retrieval-Augmented Generation (RAG) for factual grounding. The model is trained on student-contributed data, open educational resources (OER), LLM-generated QA examples, and public benchmarks. Evaluation is conducted using held-out student annotations and test splits from restricted datasets.

## 2 Data

Our training data comprises (1) student-generated preference pairs, (2) open educational resource (OER) corpora, (3) LLM-generated MCQA and reasoning data, and (4) selected public datasets. Evaluation is conducted on held-out student data and reserved test splits from approved benchmarks.

**Student-Contributed Preference Data.** We use EPFL-style preference pairs from Milestone 1 to train the DPO model.

**Open Educational Resources.** We preprocess STEM content from sources like EPFL GitHub, OpenStax[1], MIT OCW[2], and Wikipedia[3]. Documents are deduplicated, chunked into 500-token passages, indexed for retrieval, and used as prompts for LLM data generation.

**LLM-Generated Data.** We use LLMs to generate (1) MCQA pairs for training MCQA and RAG models, and (2) Chain-of-Thought (CoT) explanations to improve reasoning. QA examples are generated from OER prompts, filtered for coherence, and tokenized for supervised training. CoTs are created by prompting the model to solve each question step-by-step and are included only if multi-step reasoning is detected. For RAG, a subset of MCQA questions is paired with their source document and correct answer to form question–document–answer triplets.

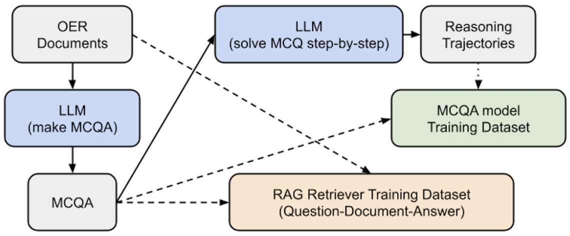


Figure 1: Pipeline for generating MCQA and RAG retriever training datasets.

| Usage | Dataset | Role |
|---|---|---|
| DPO | UltraFeedback (Cui et al., 2023) | Train |
| | HH-RLHF (Bai et al., 2022) | Train, Eval |
| | SHP (Ethayarajh et al., 2022) | Train, Eval |
| | WebGPT (Nakano et al., 2021) | Train, Eval |
| | HC3 (Guo et al., 2023) | Eval |
| MCQA | OpenBookQA (Mihaylov et al., 2018) | Train, Eval |
| | SciQ (Johannes Welbl, 2017) | Train, Eval |
| | QASC (Khot et al., 2020) | Train, Eval |
| RAG | NaturalQuestions (Kwiatkowski et al., 2019) | Train, Eval |
| | HotpotQA (Yang et al., 2018) | Train, Eval |
| | KILT (Petroni et al., 2021) | Train, Eval |
| | Aristo Tuple KB (Dalvi Mishra et al., 2017) | Retrieval |

Table 1: Public datasets used for DPO, MCQA, and RAG.

**Public Datasets.** Table 1 lists the public datasets used. Test splits from restricted benchmarks (Appendix A) are reserved for evaluation. Open-domain corpora are filtered for STEM content.

[1] https://openstax.org
[2] https://ocw.mit.edu
[3] https://en.wikipedia.org

We explore dataset combinations to optimize each model variant. For quantization, we reuse MCQA datasets.

## 3 Training

We fine-tune four task-specific models—DPO, MCQA, quantized, and RAG—based on the `Qwen/Qwen3-0.6B-Base` (Team, 2025) model.

**DPO Model.** We train the model using DPO (Rafailov et al., 2024), which aligns a target policy $\pi_\theta$ without reinforcement learning or reward modeling. Given a prompt $x$ and preference pair $(y_w, y_l)$, the loss is: $\mathcal{L}_{DPO} = -\log\sigma\big(\beta\big[\log\pi_\theta(y_w|x) - \log\pi_\theta(y_l|x) - \log\pi_{\text{ref}}(y_w|x) + \log\pi_{\text{ref}}(y_l|x)\big]\big)$, where $\pi_{\text{ref}}$ is a fixed reference, $\beta > 0$ is a temperature, and $\sigma$ is the sigmoid function.

Besides the standard DPO setup, we also plan to explore: (1) Pre-filtering (Morimura et al., 2024), removing pairs with weak preferences where $\Delta R = R(y_w) - R(y_l) < \tau$; (2) Pre-DPO (Pan et al., 2025), replacing $\pi_{\text{ref}}$ with an intermediate $\pi_{\text{pre}}$ that is periodically updated to stabilize early training.

**MCQA Model.** We fine-tune the MCQA model using supervised learning on multiple-choice question-answer pairs. The model is trained to maximize the likelihood of the correct choice: $\mathcal{L}_{\text{MCQA}} = -\log\pi_\theta(y_{\text{correct}} \mid x)$, where $x$ denotes the question, and $\pi_\theta$ is the model's output distribution over answer options.

We also plan to explore: (1) CoT distillation (Li et al., 2024; Wang et al., 2023), where rationales generated by a teacher LLM are used to guide the student model during training; (2) Permutation-based debiasing (Liusie et al., 2024; Zheng et al., 2024), which mitigates option-position bias by randomly shuffling answer choices during training and evaluation.

**Quantized Model.** We experiment with post-training quantization using GPTQ (Frantar et al., 2023) and quantization-aware training via QLoRA (Dettmers et al., 2023). We also consider recent QAT extensions such as LR-QAT (Bondarenko et al., 2024) and DL-QAT (Ke et al., 2024), which enhance efficiency through low-rank reparameterization and weight decomposition.

**RAG Model.** We implement a standard RAG pipeline with a document encoder, query encoder, and generation model. Training is performed on question–document–answer triplets constructed from our dataset. For the retriever, we will experiment with encoder setups including dense models such as BGE (Xiao et al., 2023), as well as hybrid dense–sparse retrieval methods.

Final models are selected based on validation performance on held-out data.

## 4 Evaluation

**DPO Model.** We compute preference accuracy on held-out student preference pairs and public datasets with human preference annotations. As an optional extension, we use GPT-4-as-a-judge (OpenAI et al., 2024) to compare model outputs on a subset of prompts. We use the preference accuracy of the base `Qwen3-0.6B` model as our baseline.

**MCQA Model.** We report multiple-choice accuracy on held-out synthetic MCQA data and public benchmarks, including test splits from restricted datasets listed in Appendix A. Baseline performance is taken from the `Qwen3-0.6B` model fine-tuned without CoT or debiasing.

**Quantized Model.** We evaluate quantized models on the same MCQA datasets to assess accuracy retention. We report accuracy degradation relative to the full-precision MCQA model.

**RAG Model.** We evaluate retrieval with Recall@k and Mean Reciprocal Rank (MRR), and generation with Exact Match (EM), F1, ROUGE-L, and BLEU. We compare performance against a RAG baseline trained without encoder tuning and without question–document alignment.

## 5 Roles

Each team member will be responsible for one model. Table 2 shows the specific assignments.

| Model | Student |
|---|---|
| DPO | Kyuhee |
| MCQA | Shuli |
| Quantized | Clementine |
| RAG | Stergios |

Table 2: Student roles for the project

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## A Restricted Datasets

To prevent test data contamination, the following datasets are strictly reserved for evaluation purposes only.

- MMLU (Hendrycks et al., 2021)
- MMLU-Pro (Wang et al., 2024)
- GPQA (Rein et al., 2023)
- INCLUDE (Romanou et al., 2024)
- NLP4Education (Borges et al., 2024)
- AI2 ARC (Clark et al., 2018)
- TheoremQA (Chen et al., 2023)
- MuSR (Sprague et al., 2024)
- GSM8K (Cobbe et al., 2021)
- AIME (Di Zhang, 2025)